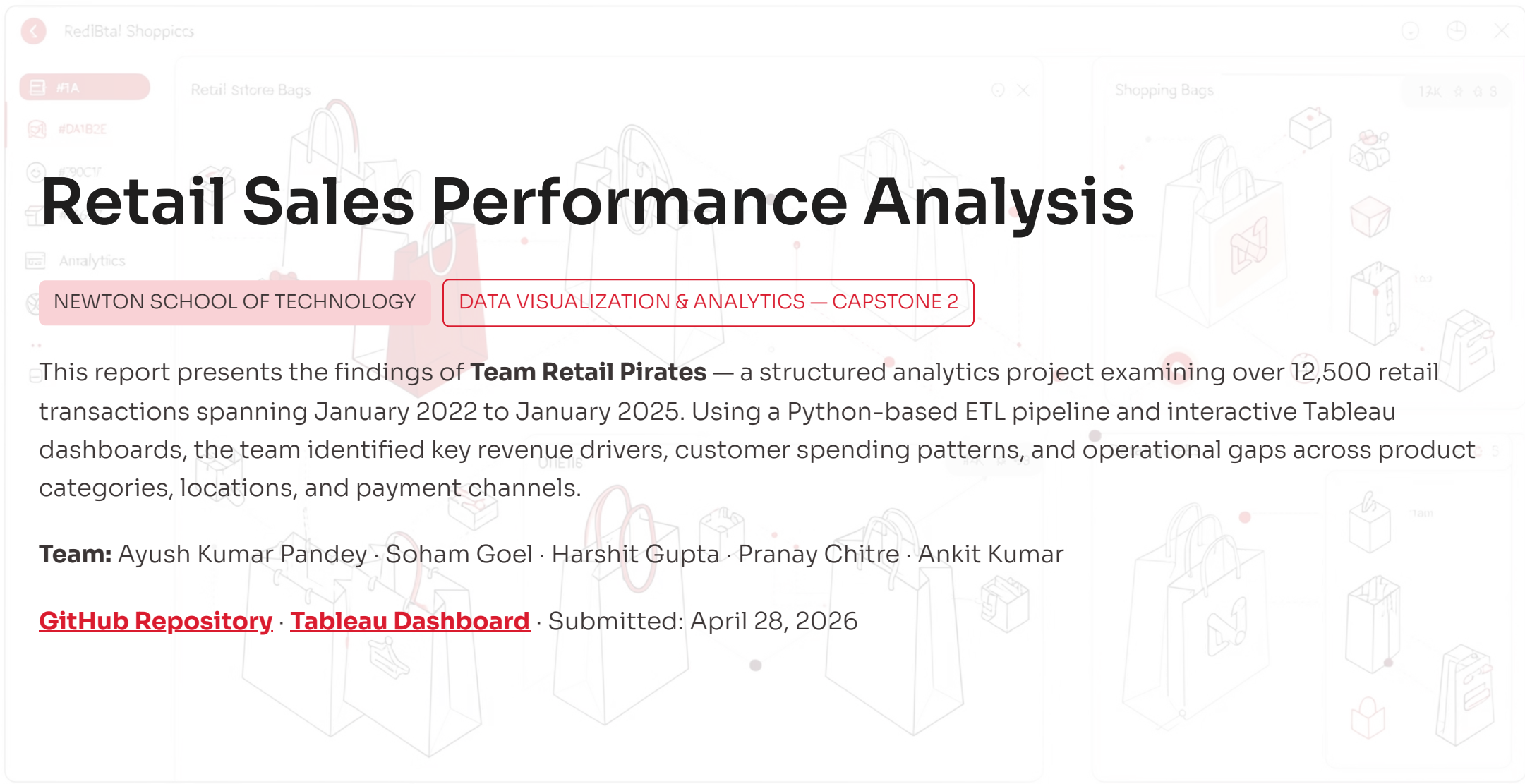




Retail Sales Performance Analysis

NEWTON SCHOOL OF TECHNOLOGY

DATA VISUALIZATION & ANALYTICS — CAPSTONE 2

This report presents the findings of **Team Retail Pirates** — a structured analytics project examining over 12,500 retail transactions spanning January 2022 to January 2025. Using a Python-based ETL pipeline and interactive Tableau dashboards, the team identified key revenue drivers, customer spending patterns, and operational gaps across product categories, locations, and payment channels.

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[GitHub Repository](#) · [Tableau Dashboard](#) · Submitted: April 28, 2026

Executive Summary

The Problem

Retail businesses generate large volumes of transactional data daily, yet few use it to drive better decisions. Sales performance across categories, locations, and customer segments remains largely invisible to decision-makers, leading to uninformed pricing, inventory, and promotional choices.

Our Approach

Transaction data covering 12,575 records was processed through a three-stage Python pipeline: extraction, cleaning, and feature engineering. Missing values were imputed, 157 outliers removed via IQR filtering, and nine derived columns added. The cleaned data was connected to Tableau to build two interactive dashboards — an Executive Summary view and an Operational Drill-Down view.

Revenue Concentration

A small number of product categories generate a disproportionate share of total revenue, creating structural dependency risk.

Discount Data Gap

~33% of discount records were missing, making it impossible to measure promotional ROI accurately.

High-Value Buyers

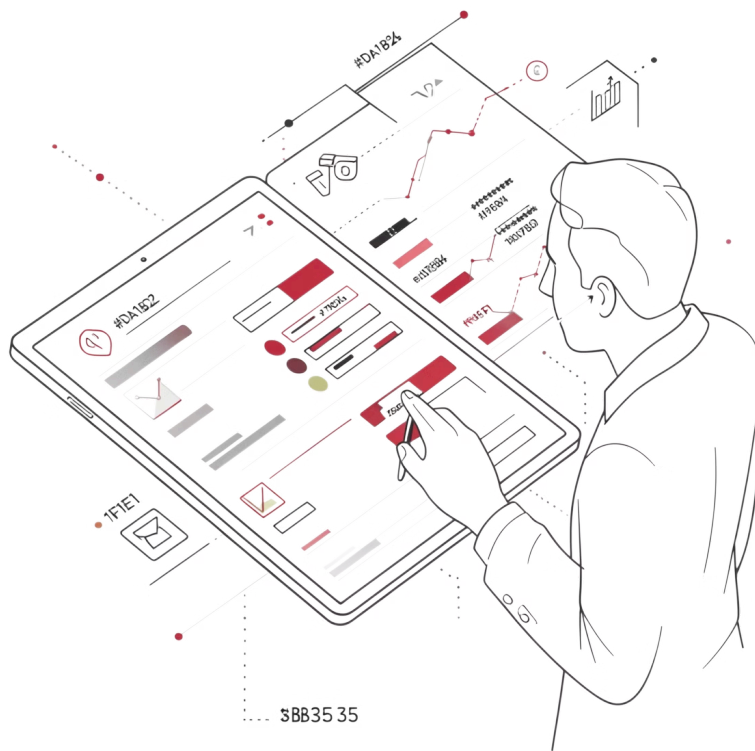
Customer spending is concentrated in a small high-value segment that drives a significant share of total revenue.

Payment Preferences

Payment method preferences differ by location, suggesting channel-specific strategies could improve conversion rates.

Sector Context & Business Value

The retail sector is one of the largest and most data-intensive industries globally. Retailers collect transaction records at scale — covering product categories, pricing, discounts, payment channels, and geographic locations. The challenge is not in collecting data, but in organising and interpreting it to make decisions that affect revenue, inventory, and customer satisfaction.



Who This Is For

This project is designed for **retail operations managers** and **senior business analysts** who need to understand sales trends across categories, regions, and customer types. The dashboards support decisions around pricing, inventory management, and promotional planning.

Why This Problem

Retail sales data is widely available and directly tied to business outcomes. The multi-year dataset (2022–2025) allows for meaningful trend analysis, offering a clear path from raw transactional data to actionable business insight.

Business Value Delivered

- Visibility into where revenue is coming from and which segments are most valuable
- Identification of discount tracking gaps and their impact on margin
- Support for targeted inventory and marketing decisions

Dataset at a Glance

12,575

Total Transactions

Records spanning the full analysis period

3

Years Covered

January 2022 to January 2025

8

Product Categories

Including Electronics, Clothing, and Grocery

25

Unique Customers

Note: unusually low — likely a simulated dataset

Payment Methods

- Cash
- Credit Card
- Digital Wallet

Purchase Channels

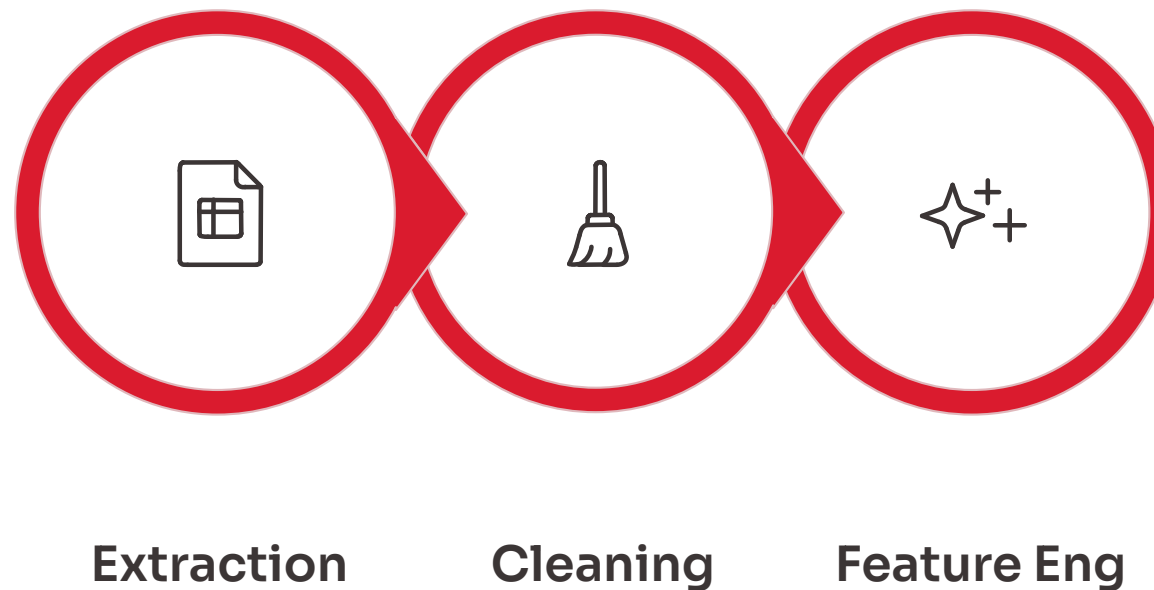
- Online
- In-Store

Key Data Limitations

- ~33% of Discount Applied values were missing (4,199 out of 12,575 records)
- Only 25 unique customers — not representative of a real retail population
- No demographic data (age, gender, income) included
- No cost-of-goods data — profit margin analysis is not possible

Data Cleaning & ETL Pipeline

All cleaning and transformation steps were carried out in Python using Jupyter Notebooks, committed to the GitHub repository under the `notebooks/` directory. The pipeline followed a strict three-stage process to ensure data quality and analytical readiness.



The cleaned dataset was exported to `data/processed/cleaned_data.csv` and used as the primary source for all subsequent EDA, statistical analysis, and Tableau dashboard connections.

Cleaning Highlights

- Duplicate rows identified and removed
- Numeric columns: median imputation
- Categorical columns: mode imputation
- 157 rows removed via IQR outlier filtering
- Transaction Date converted to datetime format

Key Assumptions

- Extreme values in `price_per_unit` and `quantity` were treated as data errors rather than genuine high-value transactions
- Mode imputation for `Discount Applied` assumes the most common discount status is representative of missing cases

KPI & Metric Framework

Nine key performance indicators were defined to measure sales performance across all analytical dimensions. All KPIs are computed in Python and referenced directly in Tableau.

KPI	Definition	Business Relevance
Total Revenue	Sum of total_spent across all transactions	Measures overall sales performance
Average Order Value (AOV)	Mean of total_spent per transaction	Indicates typical customer spend per visit
Revenue by Category	total_spent grouped by category	Identifies best and worst performing product lines
Revenue by Location	total_spent grouped by location	Highlights high-performing regions or stores
Discount Rate	Percentage of transactions with discount applied	Tracks promotional activity and margin impact
Payment Method Split	Count and revenue share by payment method	Informs payment channel strategy
Spending Segment Distribution	Count of customers in Low / Medium / High tiers	Supports customer tiering and targeting
Monthly Revenue Trend	total_spent aggregated by year and month	Tracks growth or decline over time
Bulk Purchase Rate	Percentage of transactions flagged as is_bulk	Identifies bulk-buying behaviour patterns

Exploratory Data Analysis



Revenue Over Time

Monthly revenue aggregated from 2022–2025 revealed seasonal patterns. Quarters with significant dips or spikes were flagged for deeper investigation against category and payment method data.



Customer Spending Distribution

The `spent_segment` column revealed a large concentration in the Low tier, suggesting average transaction size has room to grow through upselling or bundle strategies.



Category & Location Performance

Revenue broken down by eight product categories and by location identified which lines drive the most revenue and which locations underperform relative to transaction volume.



Payment Method Analysis

Transaction counts and revenue split by Cash, Credit Card, and Digital Wallet showed that payment preferences differ by location — informing checkout infrastructure decisions.

Top 10 Analytical Insights

Revenue Concentration Risk

A small number of product categories generate a majority of revenue — a disruption in those lines would have an outsized impact on overall performance.

Discount Data Blind Spot

~33% of discount records missing means the business cannot measure whether its discounting strategy is working.

High-Spend Customer Dependency

High-spending customers represent a small fraction of the base but account for a large share of revenue — losing even a few would meaningfully affect totals.

Payment Friction by Location

Payment method usage differs by location, suggesting some stores are not offering the options customers prefer — potentially contributing to abandoned purchases.

Day-of-Week Revenue Peaks

Certain days consistently show higher transaction volumes. If staffing and promotions are not aligned with these peaks, revenue is left on the table.

Bulk Buyer Opportunity

Bulk transactions occur regularly and carry higher spend per transaction, yet this segment remains under-served.

Pricing Inconsistency Across Categories

Revenue per unit varies significantly — low revenue-per-unit, high-volume categories represent price optimisation opportunities.

Seasonal Revenue Exposure

Quarterly variability suggests seasonal exposure. Without structured responses to slower quarters, revenue gaps go unaddressed.

Low-to-Medium Segment Upsell Potential

This segment makes up the majority of transactions. Even a modest increase in average order value through bundling would produce measurable gains.

Dataset Generalisability Limits

Only 25 distinct customers limits how confidently trends can be generalised to a larger retail population.

Recommendations & Expected Impact

#	Recommendation	Action	Expected Impact
1	Fix Discount Data Capture	Implement mandatory discount flag at POS	Enables accurate promotional ROI; prevents revenue leakage
2	High-Spend Customer Retention	Loyalty programme with personalised offers for High segment	Protects 3–4% of annual revenue via improved retention
3	Category Supply Prioritisation	Prioritise stock levels for top-performing categories	Recover ~1% of total revenue lost to stockouts
4	Payment Method Alignment	Audit and align payment options by location	Recover 5–10% of abandoned transactions in affected areas
5	Pricing Review for High-Volume Categories	Review pricing for low revenue-per-unit categories	A 5% price improvement directly adds 5% to those lines' revenue

 These estimates are illustrative. Precise figures require cost-of-goods, margin data, and customer elasticity information not present in this dataset.

Tableau Dashboards & Project Conclusion

Dashboard 1: Executive Summary

High-level snapshot for senior stakeholders — total revenue, AOV, revenue by category, and monthly trend lines. Designed for a quick read of business performance across key dimensions.

[View Executive Summary →](#)

Dashboard 2: Operational Drill-Down

Analyst view with filters for location, category, payment method, and time period. Supports investigation of specific performance questions such as which locations underperform in a given quarter.

[View Operational Drill-Down →](#)

Category Filter

Select one or more product categories

Location Filter

Filter all views by region or store

Date Range Selector

Restrict view to a defined time period

Payment Method Toggle

Isolate transactions by payment type

This project demonstrates how a structured analytics pipeline can turn raw transactional data into decisions that protect and grow revenue. Acting on the recommendations — particularly around discount data capture, high-spend customer retention, and category supply prioritisation — would give the business clearer visibility into its performance and concrete levers to improve it.